Small Models that Forget: Exploiting Temporal Locality with Online Learning at Inference Time

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Abstract

Many inputs that machine learning models see at inference time have strong temporal locality: consecutive inputs tend to be similar and come from a small part of the input distribution. A model facing such inputs does not need to keep everything it knows ready at all times. It may be enough to remember some basic knowledge and adapt quickly to the current context. We study whether this allows smaller models. Multilayer perceptrons of decreasing size are trained offline on MNIST and then updated online, one image at a time, while they classify a stream of test images arranged in bursts of three digits, with 10% intruders from outside the current burst. A shuffled copy of the same stream serves as a control without locality. With plain SGD, small models trained online classify the bursty stream more accurately than their frozen offline version, and the gain over the shuffled stream grows as the model shrinks and as the online learning rate increases. For the smallest model (6.2 KB), online learning raises accuracy from 64.0% to 69.0%, while on the shuffled stream it stays at 64.0%. The price is paid on the intruders, and it grows with the learning rate. Splitting the weights into slow and fast parts helps when the fast part is only the output bias, which raises the smallest model to 71.5%, but a running label prior that learns no weights does as well or better. In this stream the locality is only in the labels, and every method that exploits it loses accuracy on the intruders.

1 Introduction

A trained network is usually deployed frozen: it must classify any input from the whole distribution it was trained on. In many applications, however, the inputs arrive in a stream with strong temporal locality. A camera in a kitchen sees kitchen objects for hours; a speech recognizer hears the same speaker for a whole conversation; a sensor stays in the same regime for long periods. At any moment only a small part of the input distribution is active.

Computer memory hierarchies have relied on this property for decades: a small cache that holds the recently used data serves most requests [2]. We ask whether the same idea applies to the knowledge stored in a network. A model does not need to remember everything all the time if its inputs have strong temporal locality. It may only need some basic knowledge plus the ability to adapt fast to the current context.

Our hypothesis is that, under severe capacity limits, online training at inference time can outperform the same model trained offline and then frozen. A frozen small model must spread its limited capacity over the whole distribution. An online model can spend it on the part of the distribution that is active now. The price is forgetting: neural networks trained sequentially lose what they learned earlier, a behavior known as catastrophic forgetting [11, 3]. This recency bias is usually treated as a problem. In a stream with temporal locality it may be useful, because what was seen

recently is what will be seen next. Real streams are not perfectly local, though, and a model that adapts to the current context may fail on the inputs that do not fit it.

This paper reports a first set of experiments on MNIST [9]. Experiment 1 checks whether small networks exploit temporal locality when they keep learning online with plain SGD, and how the online learning rate matters. Experiment 2 splits knowledge by time scale, with slow weights and fast weights that adapt to the current context and then fade, and compares it with a running label prior.

2 Related work

Catastrophic forgetting and continual learning. Continual learning tries to learn a sequence of tasks without forgetting the earlier ones [12]. Elastic weight consolidation [7], for example, protects the weights that matter for old tasks. Our setting reverses the goal: we accept forgetting and ask whether it can be used.

Two time scales. Hinton and Plaut [5] added to each connection a fast weight that learns quickly and decays, on top of a slow weight, to recover old memories after interference. Complementary learning systems theory [10] explains the division between a fast-learning hippocampus and a slow-learning neocortex by the need to learn new experiences without overwriting old knowledge. Fast weights in recurrent networks [1] store recent hidden states in a decaying outer-product memory. Experiment 2 uses the formulation of [5], with the fast part meant to track the current context and forget it.

Test time adaptation. Several methods update a model while it is used, for example with a self supervised task [13] or by minimizing the entropy of the predictions [14]. They target distribution shift and do not use labels. Dynamic evaluation [8] updates a language model on the recent part of the sequence it is predicting, which exploits local regularities in text, the closest idea to ours.

Model compression. Compression reduces model size by pruning, quantization or distillation [4], keeping the model frozen. We study a different route to small models that is complementary to compression.

3 Method

3.1 Input streams

The stream contains 10,000 positions built from MNIST test images. It is divided into 193 bursts of random length between 10 and 95 images (51.8 on average). Each burst picks 3 distinct digits at random, and every position in the burst is a random test image of one of those digits, drawn with replacement. Then 10% of the positions (1,000), chosen at random, are replaced by *intruders*: a random test image of a digit that does not appear in that burst. The other 9,000 positions are the *regular* images. The stream holds 6,299 distinct test images. As a reference for how much locality the stream has, predicting the label of the previous image is correct 26.3% of the time.

The control stream is a random permutation of the same positions. It contains exactly the same images, so any difference in accuracy comes from their order. On the control stream, predicting the previous label is correct 10.3% of the time.

The images of a burst are drawn independently given their digit, so the locality of the stream is in the labels only.

3.2 Protocol

Each model is first trained offline on the 60,000 MNIST training images. We call this the frozen model. Then it processes a stream one image at a time. For each image, the model first predicts its label, and the prediction counts towards the online accuracy. Then it takes one optimization step on that single image with its true label. This assumes that the label becomes available after each prediction, as in prequential evaluation. Each stream starts from the same frozen weights.

We report these quantities, all on the same 10,000 positions:

- *frozen accuracy*: the frozen model without updates (equal on both streams, since they hold the same images);
- *online accuracy on bursts* and *online accuracy on random*: the online model on each stream;
- *online gain*: online accuracy on bursts minus frozen accuracy;
- *locality effect*: online accuracy on bursts minus online accuracy on random;
- the online gain restricted to the regular images and to the intruders, each against the frozen accuracy on the same images.

Differences are given in percentage points (pp). The online gain tells whether online learning is worth it on the bursty stream. The locality effect isolates the part of the gain that comes from temporal locality. It removes the effect of simply training more, and of seeing some test images more than once, which happens in both streams. The gain on intruders measures the cost of adapting to the current context.

Model size is the memory needed to store all parameters and buffers as 32 bit floats. All runs use a fixed seed (0).

3.3 Models

The multilayer perceptron (MLP) has one hidden layer with ReLU: $784 \to h \to 10$, with h between 128 and 2. It is trained offline with Adam [6], learning rate 0.001, batch size 128, 10 epochs. The online phase uses plain SGD without momentum.

4 Experiments

4.1 Experiment 1: online learning rate

The first experiment updates MLPs with $h \in \{128, 16, 8, 4, 2\}$ online with SGD at four learning rates: 0.0003, 0.001, 0.003 and 0.01. Table 1 gives the online gain and the locality effect, and Table 2 splits the gain into regular images and intruders. Figures 1 and 2 show the results against model size.

With SGD and learning rates up to 0.001, online learning improves on the frozen model at every size. The online gain grows as the model shrinks: about 0.2 pp for $h \geq 16$, and 5.00 pp for $h = 2$ at learning rate 0.001 (from 63.95% to 68.95%). On the shuffled stream the same small model stays where it was (63.96%), so the gain comes from the order of the inputs and not from the extra

Table 1: Experiment 1: MLPs with online SGD at four learning rates. Online gain (bursts minus frozen) and locality effect (bursts minus random), in pp.

			online gain				locality effect			
h	KB	frozen	0.0003	0.001	0.003	0.01	0.0003	0.001	0.003	0.01
128	397.5	97.82	+0.22	+0.17	−0.24	−2.59	−0.05	−0.11	+0.03	+0.68
16	49.7	95.07	+0.16	+0.22	−0.60	−5.37	+0.00	+0.15	+0.30	+1.97
8	24.9	92.46	+0.62	+1.00	+0.03	−4.42	+0.36	+1.18	+1.99	+3.50
4	12.5	85.68	+1.02	+1.12	−0.51	−8.33	+1.41	+2.60	+3.64	+8.05
2	6.2	63.95	+3.41	+5.00	+4.51	−6.90	+1.91	+4.99	+9.05	+9.71

Table 2: Experiment 1: online gain on the bursty stream, restricted to the 9,000 regular images and to the 1,000 intruders, each against the frozen model on the same images, in pp.

	regular images				intruders			
h	0.0003	0.001	0.003	0.01	0.0003	0.001	0.003	0.01
128	+0.21	+0.14	−0.19	−2.34	+0.30	+0.40	−0.70	−4.90
16	+0.16	+0.37	−0.16	−4.27	+0.10	−1.20	−4.60	−15.30
8	+0.71	+1.40	+0.83	−3.03	−0.20	−2.60	−7.20	−17.00
4	+1.80	+2.60	+1.58	−5.27	−6.00	−12.20	−19.30	−35.90
2	+4.41	+6.98	+7.13	−3.98	−5.60	−12.80	−19.10	−33.20

training. Learning rate 0.001 gives the largest online gain at every size except $h = 128$, where 0.0003 is ahead by 0.05 pp.

The locality effect grows in two directions: as the model gets smaller and as the learning rate gets larger. At learning rate 0.01 it reaches 9.71 pp for $h = 2$, but there the online model is worse than the frozen one on both streams, even on the regular images: the effect is large because accuracy on the shuffled stream collapses faster.

The intruders show the cost of adapting. For the small models, the faster the learning rate, the more accuracy is lost on intruders: for $h = 2$, from −5.6 pp at 0.0003 to −33.2 pp at 0.01. At $h = 2$, learning rate 0.003 gains as much as 0.001 on the regular images (+7.13 against +6.98 pp) but loses more on intruders (−19.1 against −12.8 pp), which puts it behind overall. The large model ($h = 128$) shows almost no locality effect and no cost on intruders at learning rates up to 0.001. With enough capacity, the frozen model already covers the whole distribution and there is little to gain from adapting to the current context.

4.2 Experiment 2: two time scales

The online loss only sees the current image, so plain online SGD has to trade adapting to the current burst against keeping what it knows. The second experiment splits knowledge by time scale [5]. Each chosen parameter gets a slow part S and a fast part F, and the network uses $S + F$. After each image, with g the gradient of the loss on that image,

$$S \leftarrow S - \eta_{\text{slow}}\, g,$$
$$F \leftarrow \lambda F - \eta_{\text{fast}}\, g,$$

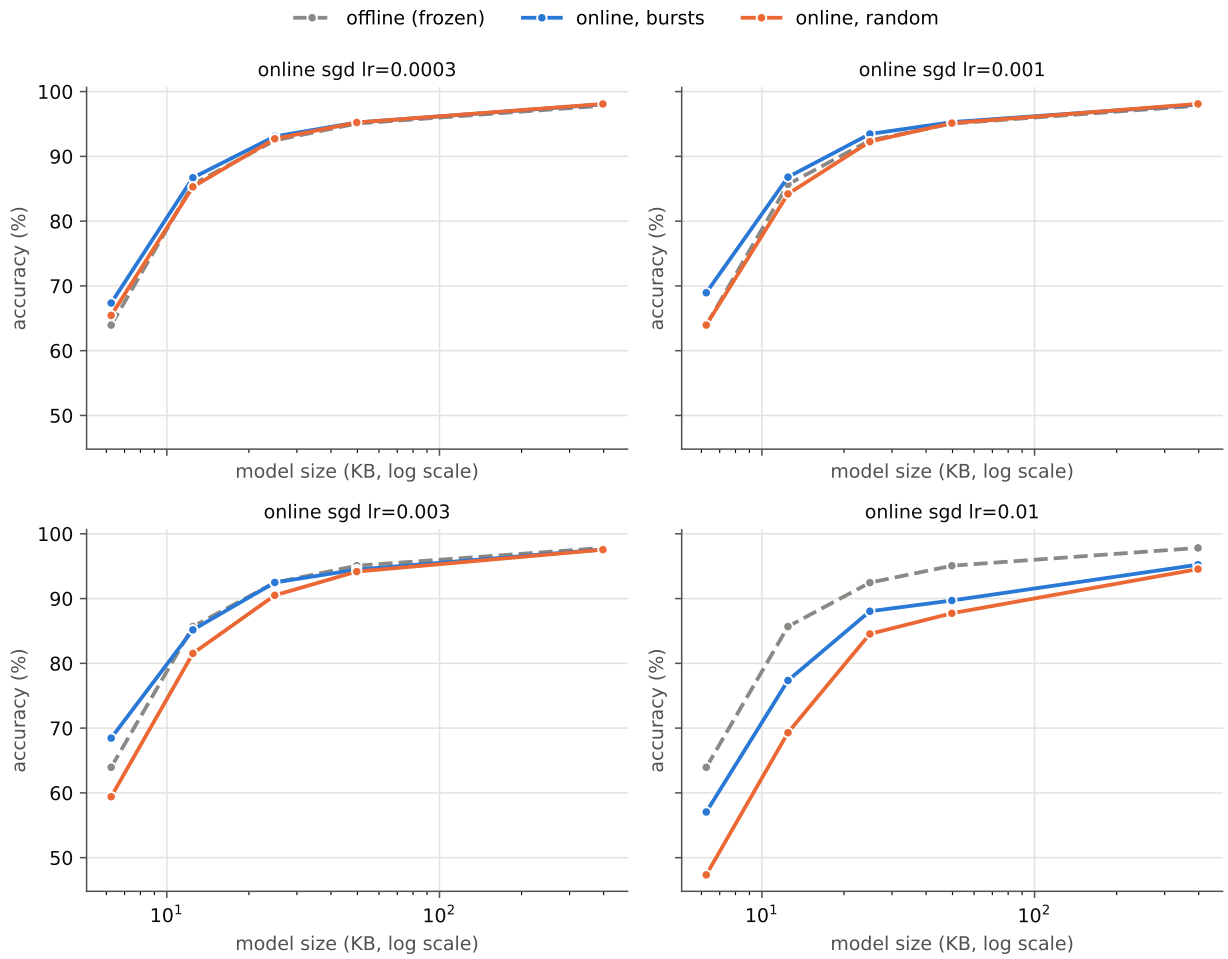


Figure 1: Experiment 1: accuracy against model size, one panel per online learning rate.

with F starting at zero. The fast part is a decaying sum of recent gradients: it follows the current burst and forgets it within about $1/(1-\lambda)$ images.

We compare four online variants, on MLPs with $h \in \{128, 8, 4, 2\}$:

- *sgd*: plain SGD, $\eta_{\text{slow}} = 0.001$, no fast part (the best setting of Experiment 1);
- *fast bias*: $\eta_{\text{slow}} = 0.001$, with a fast part on the output bias only ($\eta_{\text{fast}} = 1$, $\lambda = 0.97$), which adds 10 values;
- *fast all*: $\eta_{\text{slow}} = 0.001$, with a fast part on every weight and bias ($\eta_{\text{fast}} = 0.003$, $\lambda = 0.99$), which doubles the model;
- *label prior*: frozen weights and a running estimate p of the label frequencies, $p \leftarrow 0.9\,p + 0.1\,\text{onehot}(y)$ after each prediction, starting uniform and kept $\geq 10^{-3}$. Prediction adds $\log((1-m)\,p + m/10)$ to the logits. The mix m keeps digits that are absent from the recent past possible; $m = 0.2$ gave the best accuracy among $\{0, 0.05, 0.1, 0.2\}$ for $h = 2$. It learns no weights and adds 10 values.

The label prior is a baseline that exploits exactly the locality in the labels and nothing else. The fast-part hyperparameters were tuned on the stream without intruders, and m on the stream with

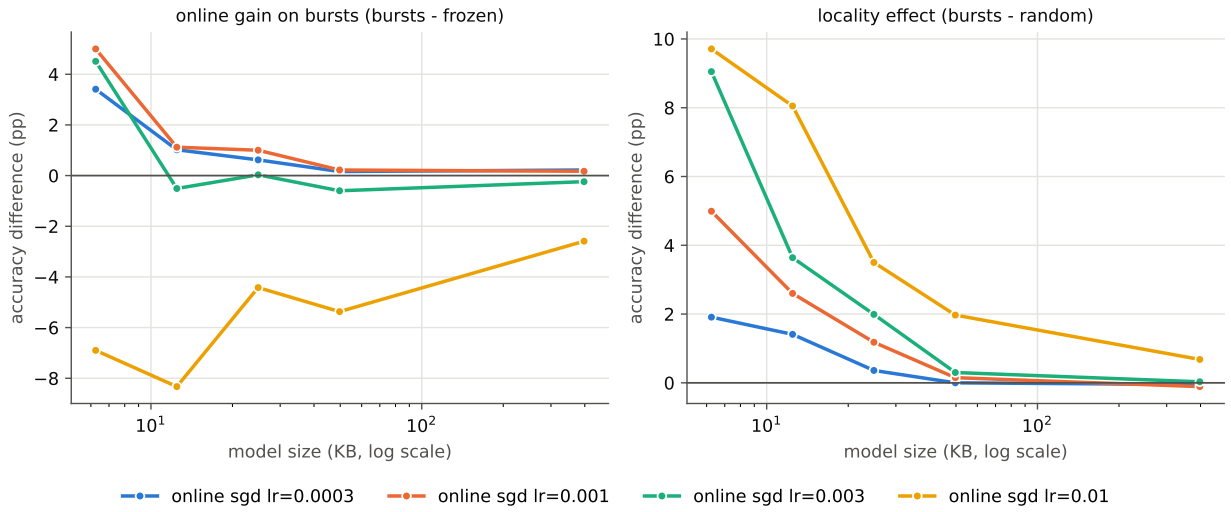


Figure 2: Experiment 1: online gain (left) and locality effect (right) against model size, one line per online learning rate.

Table 3: Experiment 2: online gain on the bursty stream for the four variants, overall and restricted to the regular images and to the intruders, each against the frozen model on the same images, in pp. The oracle restricts the frozen model to the digits of the current burst.

	overall				regular images					intruders			
h	sgd	prior	f. bias	f. all	sgd	prior	f. bias	f. all	oracle	sgd	prior	f. bias	f. all
128	+0.17	+0.42	+0.31	−0.03	+0.14	+0.60	+0.31	+0.05	+1.41	+0.4	−1.2	+0.3	−0.8
8	+1.00	+2.20	+1.28	+0.29	+1.40	+2.89	+1.70	+1.05	+4.77	−2.6	−4.0	−2.5	−6.6
4	+1.12	+3.22	+1.56	−0.31	+2.60	+5.00	+3.51	+1.74	+9.02	−12.2	−12.8	−16.0	−18.8
2	+5.00	+8.78	+7.51	+4.25	+6.98	+12.90	+11.42	+6.84	+21.55	−12.8	−28.3	−27.7	−19.1

them. As a ceiling for the regular images, we also score an oracle: the frozen model with its prediction restricted to the digits of the current burst, which it knows in advance. It is always wrong on intruders, by construction.

Table 3 gives the results, and Figure 3 shows the accuracy on intruders and on regular images against model size.

No variant gains on the regular images without losing on the intruders. The label prior gains most on the regular images at every size (+12.90 pp for $h = 2$, 60% of the oracle's +21.55) and has the best overall accuracy at every size. A fast output bias is second. For $h = 128$ and $h = 8$ it loses about the same on intruders as plain SGD while gaining more on the regular images, a clear improvement; for $h = 4$ and $h = 2$ it loses much more (−27.7 against −12.8 pp for $h = 2$). Plain SGD gains least of the three but has the smallest cost on intruders for the small models. Making the whole network fast is worse than plain SGD at every size, with less gain on the regular images and more loss on the intruders, and it doubles the model.

On the shuffled stream, the label prior and the fast variants lose 5.5 to 6.7 pp for $h = 2$, while plain SGD loses nothing.

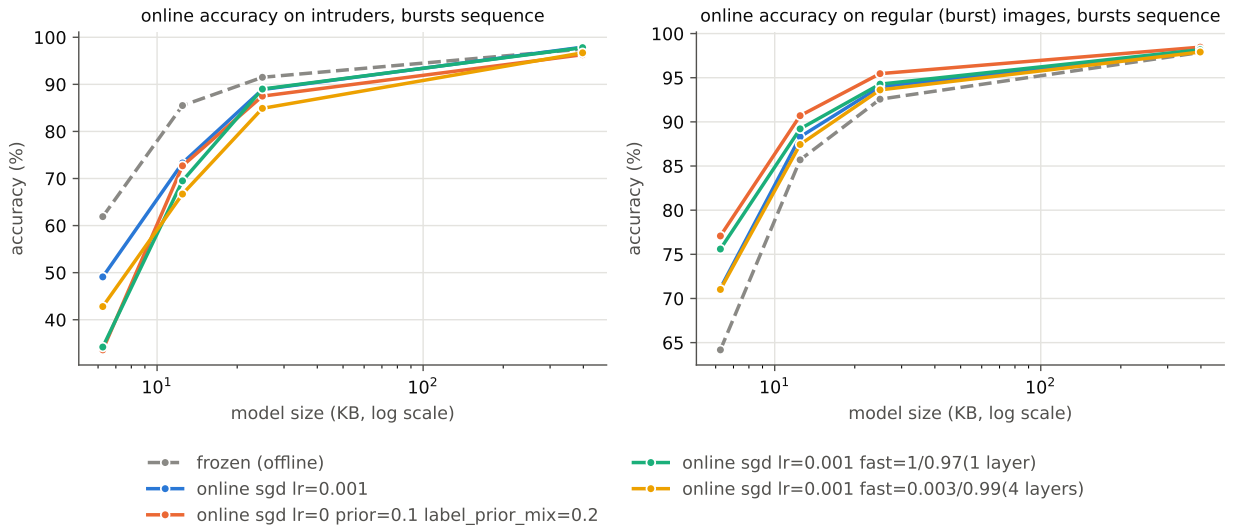


Figure 3: Experiment 2: online accuracy on the intruders (left) and on the regular images (right) of the bursty stream against model size, one line per variant, with the frozen model dashed.

5 Discussion

The experiments support the hypothesis for small MLPs. When capacity is very limited, a model that keeps adapting to the current context classifies a stream with temporal locality better than the same model frozen, and the advantage comes from the locality: it disappears or reverses on a shuffled copy of the stream. The recency bias of online training, which hurts on the shuffled stream, helps on the bursty one. Seen this way, the learning rate controls the length of the model's memory: a larger rate forgets faster and adapts faster. The best value should depend on how long the contexts last, which here is about 50 images.

The intruders make the trade-off visible. Every way of exploiting the burst, from a faster learning rate to fast weights or a label prior, also makes the model more confident in the current digits and costs accuracy on the inputs that do not belong to the burst. How much this matters depends on the application: the cost of an error on an unexpected input may be much higher than on a common one.

The strength of the label prior says something about the stream rather than about fast weights. The images of a burst share only their labels, so the most useful thing to remember about the recent past is which digits are active. A fast output bias can learn this, and that is what it does best; a fast copy of the whole network has more to learn and gains less. A stream whose locality is also in the inputs would be a fairer test of fast weights, because a label prior cannot exploit it.

The results also have clear limits. The task is MNIST and the stream is synthetic, with an idealized form of locality: bursts with sharp boundaries and only three classes each, and intruders spread uniformly. The online model receives the true label right after each prediction, which many applications cannot provide. Every result comes from a single run with seed 0, so differences below about 0.5 pp may not be meaningful, and accuracies on the 1,000 intruders have a sampling uncertainty of about 1.5 pp for $h = 2$. The hyperparameters of Experiment 2 were tuned on the same streams they are scored on. The stream draws test images with replacement, so the online model sees some images more than once; the shuffled control has the same repetitions, and the locality effect cancels this out, but the online gain does not.

6 Conclusion and next steps

Small MLPs trained online at inference time exploit the temporal locality of their input stream. Below some model size they become more accurate than their frozen offline version, and the advantage is tied to the order of the inputs. The effect grows as the model shrinks and as the online learning rate increases, and so does the cost on the inputs that do not fit the current context. A fast output bias on top of slow weights improves the trade-off for the larger models, but a running label prior without any learning does as well or better, because the locality of the stream is in the labels.

Next steps include streams with locality in the inputs (for example one shared distortion per burst), repeating the runs with several seeds, tuning on separate streams, varying the burst length, the number of classes per burst and the intrusion rate, testing streams with natural locality (for example video) and other architectures, and replacing the true labels with weaker feedback.

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